LUND UNIVERSITY

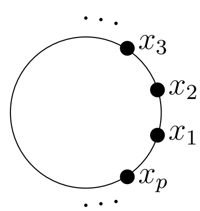
Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written Examination
Group and Ring Theory
MATP33
Wednesday, 25 May 2022
Duration: 08:00–13:00

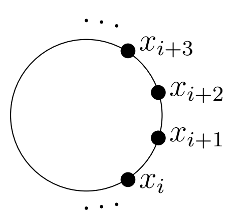
No books, notes, computational devices, etc. are allowed. Use clear handwriting and give clear careful motivations. Please write your anonymous identification number on each sheet of paper.

1. Find the number of different necklaces with p spherical beads, for p an odd prime, where the beads can have any of n different colours. Two necklaces are considered the same, if one necklace can be transformed into the other under rotations and reflections.

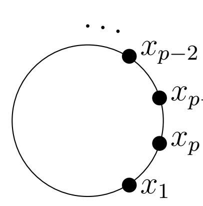
Solution. (5 marks) Let X be the set of all possible necklaces. Clearly $|X| = n^p$. Let

$x =$ 

be a necklace. Let $G = \langle a, b \mid a^p = b^2 = 1 \rangle$ be the dihedral group of order $2p$. Then X is a G-set under the action

$a^i \cdot x =$ 

and

$b \cdot x =$ 

where subscripts are modulo p.

Clearly the action of any fixed element $a^i b^\epsilon \in G$, for $i \in \mathbb{F}_p$ and $\epsilon \in \{0, 1\}$, on a given necklace yields the same necklace. (The beads are just being permuted cyclically after performing a reflection.) So the number of orbits is the same as the number of different necklaces. We now compute X_g for each $g \in G$.

First

$$X_e = \{x \in X \mid e \cdot x = x\} = X.$$

So $|X_e| = n^p$.

Let $g = a^j$ for some $j \in \mathbb{F}_p^*$. Since $\langle a \rangle$ is cyclic of prime order, the element g generates $\langle a \rangle$. Then

$$\begin{aligned} X_g &= \{x \in X \mid g \cdot x = x\} \\ &= \{x \in X \mid g^i \cdot x = x, \quad \forall i\} \\ &= \{x \in X \mid h \cdot x = x, \quad \forall h \in \langle a \rangle\}. \end{aligned}$$

Therefore X_g consists of all those necklaces which are unchanged by any permutation and these are precisely those which are of one colour. Hence $|X_g| = n$ for all $g \in \langle a \rangle \backslash \{e\}$.

For $g = b$, we see that X_b consists of all the necklaces with

$$x_1 = x_p, \quad x_2 = x_{p-1}, \quad \ldots, \quad x_{\frac{p-1}{2}} = x_{\frac{p+3}{2}}.$$

Hence $|X_b| = n^{\frac{p+1}{2}}$.

Finally let $g = a^i b$ for $i \in \mathbb{F}_p^*$. Then g has order 2. As with the above, with have $|X_g| = n^{\frac{p+1}{2}}$.

Then Burnside's Lemma then gives the number of different necklaces as

$$\begin{aligned} \frac{1}{|G|} \sum_{g \in G} |X_g| &= \frac{1}{2p} \left[n^p + n + \overset{p-1}{\cdots} + n + n^{\frac{p+1}{2}} + \overset{p}{\cdots} + n^{\frac{p+1}{2}} \right] \\ &= \frac{1}{2p} \left[n^p + n(p-1) + n^{\frac{p+1}{2}} p \right]. \end{aligned}$$

2. Let G be a group of order $351 = 3^3 \cdot 13$. Prove that G has a normal Sylow p-subgroup for some prime p dividing $|G|$.

Solution. (5 marks) By the Third Sylow Theorem, we have $n_{13} \mid 27$ and $n_{13} \equiv 1 \pmod{13}$. This forces $n_{13} \in \{1, 27\}$. If $n_{13} = 1$, we are done, so suppose that $n_{13} = 27$.

Thus there are 27 Sylow 13-subgroups of G. As 13 is prime, this forces the intersection of two distinct Sylow 13-subgroups to be trivial. Hence, the number of elements having order 13 equals $27(13-1) = 3^3(12)$.

This leaves us with 3^3 remaining elements which are not part of any Sylow 13-subgroup. Now by the Third Sylow Theorem, we know that $n_3 \geq 1$. However, no non-identity element can be part of a Sylow 3-subgroup as well as a Sylow 13-subgroup. Thus, the remaining 3^3 elements form the unique Sylow 3-subgroup which gives us that $n_3 = 1$ and we are done.

3. Let $R = M_n(F)$ be the matrix ring over a field F, and for $1 \leq i, k \leq n$, denote by e_{ik} the matrix with 1 at the (i, k) entry and 0 elsewhere. For a fixed choice of i and k, is $M = Re_{ik}$ a simple R-module?

Solution. (5 marks) Let $N \neq 0$ be an R-submodule of Re_{ik}. Note that M is the set of matrices all of whose entries are zero, except possibly in the kth column. Hence a non-zero element of N can be expressed as $\sum_{j=1}^{n} a_{jk}e_{jk}$ for $a_{jk} \in F$.

Consider $0 \neq \sum_{j=1}^{n} a_{jk}e_{jk} \in N$. Then for some ℓ, with $1 \leq \ell \leq n$, we have $a_{\ell k} \neq 0$. So $a_{\ell k}^{-1} \in F$ exists. This gives

$$e_{ik} = (a_{\ell k}^{-1} e_{i\ell}) \left(\sum_{j=1}^{n} a_{jk}e_{jk} \right) \in N,$$

as

$$e_{i\ell}e_{jk} = \begin{cases} 0 & \text{if } j \neq \ell, \\ e_{ik} & \text{if } j = \ell. \end{cases}$$

Hence $Re_{ik} \subseteq N \subseteq Re_{ik}$ giving $N = Re_{ik}$. This proves that M is simple.

4. Let M be an Artinian R-module. Show that every quotient module of M is finitely cogenerated.

Solution. (5 marks) Recall that an R-module L is said to be *finitely cogenerated* if, for each family $(L_\alpha)_{\alpha\in\Lambda}$ of submodules of L,

$$\bigcap_{\alpha\in\Lambda} L_\alpha = 0 \quad\Longrightarrow\quad \bigcap_{\alpha\in\Lambda'} L_\alpha = 0$$

for some finite subset Λ' of Λ.

Let N be a submodule of M, and consider the quotient module $L := M/N$. Assume that L is not finitely cogenerated. So there exists a family $(L_\alpha)_{\alpha\in\Lambda}$ of submodules of L with

$$\bigcap_{\alpha\in\Lambda} L_\alpha = 0 \quad \text{but} \quad \bigcap_{\alpha\in\Lambda'} L_\alpha \neq 0$$

for every finite subset Λ' of Λ. This implies that Λ must be infinite.

Let $\Lambda_1 \subsetneq \Lambda_2 \subsetneq \cdots$ be an strictly increasing chain of finite subsets of Λ, such that

$$\bigcap_{j\in\Lambda_i} L_j \supsetneq \bigcap_{j\in\Lambda_{i+1}} L_j;$$

such a chain exists, since $\bigcap_{\alpha\in\Lambda} L_\alpha = 0$. Write $K_i = \bigcap_{j\in\Lambda_i} L_j$. We have by assumption that $K_j \neq 0$ for all $j \in \mathbb{N}$, and

$$K_1 \supsetneq K_2 \supsetneq \cdots$$

is a strictly descending infinite chain of submodules of L. By the Correspondence Theorem, we obtain a strictly descending infinite chain of submodules of M, which is a contradiction to the hypothesis. Hence L is finitely cogenerated.

5. Determine the rational canonical form of

$$\begin{pmatrix} 2 & -2 & 14 \\ 0 & 3 & -7 \\ 0 & 0 & 2 \end{pmatrix}$$

considered as a complex matrix.

Solution. (5 marks) We first find the invariant factors of $A - xI$. We have

$$\begin{pmatrix} 2-x & -2 & 14 \\ 0 & 3-x & -7 \\ 0 & 0 & 2-x \end{pmatrix} \overset{C_1 \leftrightarrow C_2}{\sim} \begin{pmatrix} -2 & 2-x & 14 \\ 3-x & 0 & -7 \\ 0 & 0 & 2-x \end{pmatrix}$$

$$\overset{\frac{1}{2}R_1}{\sim} \begin{pmatrix} -1 & 1-\frac{1}{2}x & 7 \\ 3-x & 0 & -7 \\ 0 & 0 & 2-x \end{pmatrix}$$

$$\overset{C_2+(1-\frac{1}{2}x)C_1}{\sim} \begin{pmatrix} -1 & 0 & 7 \\ 3-x & (3-x)(1-\frac{1}{2}x) & -7 \\ 0 & 0 & 2-x \end{pmatrix}$$

$$\overset{C_3+7C_1}{\sim} \begin{pmatrix} -1 & 0 & 0 \\ 3-x & (3-x)(1-\frac{1}{2}x) & 14-7x \\ 0 & 0 & 2-x \end{pmatrix}$$

$$\overset{R_2+(3-x)R_1}{\sim} \begin{pmatrix} -1 & 0 & 0 \\ 0 & (3-x)(1-\frac{1}{2}x) & 14-7x \\ 0 & 0 & 2-x \end{pmatrix}$$

$$\overset{R_2-7R_3}{\sim} \begin{pmatrix} -1 & 0 & 0 \\ 0 & (3-x)(1-\frac{1}{2}x) & 0 \\ 0 & 0 & 2-x \end{pmatrix}$$

$$\overset{2R_2}{\sim} \begin{pmatrix} -1 & 0 & 0 \\ 0 & (3-x)(2-x) & 0 \\ 0 & 0 & 2-x \end{pmatrix}$$

$$\overset{C_2 \leftrightarrow C_3}{\sim} \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & (3-x)(2-x) \\ 0 & 2-x & 0 \end{pmatrix}$$

$$\overset{R_2 \leftrightarrow R_3}{\sim} \begin{pmatrix} -1 & 0 & 0 \\ 0 & 2-x & 0 \\ 0 & 0 & (3-x)(2-x) \end{pmatrix},$$

so the invariant factors of $A - xI$ are $x-2$ and $(x-2)(x-3) = x^2 - 5x + 6$. Hence the rational canonical form for A is

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & -6 \\ 0 & 1 & 5 \end{pmatrix}.$$

6. Let R be a Noetherian ring with unity. Prove that the polynomial ring $R[x]$ is also a Noetherian ring.

Solution. (5 marks) Let $\mathcal{F}$ and $\mathcal{F}'$ be the families of left ideals of R and $R[x]$, respectively. Let $n \in \mathbb{Z}_{\geq 0}$ and define a mapping $\phi_n : \mathcal{F}' \to \mathcal{F}$, where

$$\phi_n(I) = \{a \in R \mid \exists ax^n + bx^{n-1} + \cdots \in I,\ a \neq 0\} \cup \{0\}.$$

It is easy to verify that $\phi_n(I) \in \mathcal{F}$. We claim that if $I, J \in \mathcal{F}'$ with $I \subseteq J$ and $\phi_n(I) = \phi_n(J)$ for all $n \geq 0$, then $I = J$. Let $0 \neq f(x) \in J$ be of degree m. Since $\phi_m(I) = \phi_m(J)$, there exists $g_m(x) \in I$ with leading coefficient the same as that of $f(x)$, and $f(x) - g_m(x)$ is either 0 or of degree at most $m-1$. Suppose $f(x) - g_m(x) \neq 0$. As $f(x) - g_m(x) \in J$, we can similarly find $g_{m-1}(x) \in I$ such that $f(x) - g_m(x) - g_{m-1}(x) \in J$ and is either 0 or of degree at most $m-2$. Continuing like this, we arrive, after at most m steps, at

$$f(x) - g_m(x) - g_{m-1}(x) - \cdots - g_i(x) = 0,$$

where $g_m(x), g_{m-1}(x), \ldots \in I$. Then $f(x) \in I$, which proves that $I = J$, as claimed.

Let $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$ be an ascending sequence of left ideals of $R[x]$. Then for each non-negative integer n,

$$\phi_n(A_1) \subseteq \phi_n(A_2) \subseteq \phi_n(A_3) \subseteq \cdots$$

is an ascending sequence of left ideals of R. Hence there exists a positive integer $k(n)$ such that

$$\phi_n(A_{k(n)}) = \phi_n(A_{k(n)+1}) = \cdots . \tag{1}$$

Further, as R is Noetherian, the collection of left ideals $(\phi_n(A_i))$, for $n, i \in \mathbb{N}$, has a maximal element, say $\phi_p(A_q)$. Then

$$\begin{aligned} \phi_p(A_q) &= \phi_n(A_q) && \text{(for all } n \geq p) \\ &= \phi_n(A_j) && \text{(for all } n \geq p,\ j \geq q). \end{aligned}$$

Therefore, we may choose $k(n) = q$ for all $n \geq p$ in (1). Moreover, if $s = k(1) \cdots k(p-1)q$ (or even $s = \max\{k(1), \ldots, k(p-1), q\}$ works), then $\phi_n(A_s) = \phi_n(A_{s+1}) = \cdots$ for all $n \in \mathbb{N}$. Hence, by the result proved in the first paragraph, we obtain $A_s = A_{s+1} = \cdots$. Therefore $R[x]$ is Noetherian.